

CSC 215: Discrete Structures

Programming Assignment 1: Set Theory

Objective

The purpose of this assignment is to practice basic set operations through Java program.

Problem

Consider the following two sets:

$$A = \{2, 4, 6, 8, 10\}$$

and

$$B = \{6, 8, 10, 12, 14\}.$$

Write a program that stores the elements of sets A and B and performs the following set operations:

1. Union: $A \cup B$
2. Intersection: $A \cap B$
3. Difference: $A - B$
4. Difference: $B - A$

Expected Output

Your program should display the original sets and the results of each set operation. The output should be similar to the following:

A = {2, 4, 6, 8, 10}

B = {6, 8, 10, 12, 14}

A Union B = {2, 4, 6, 8, 10, 12, 14}

A Intersection B = {6, 8, 10}

A - B = {2, 4}

B - A = {12, 14}

Requirements

- Your program must correctly represent sets A and B .
- Your program must perform all four required set operations.
- Label each result in the output.
- Your program should run without errors.
- Use appropriate comments to make your code easy to understand.

Submission

Submit your complete source code file through the Canvas. Make sure your name and course information are included as comments at the beginning of your program.